



Impacts of microplastics on reproductive performance of male tilapia (*Oreochromis niloticus*) pre-fed on *Amphora coffeaeformis*

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Abstract

This study examines the potential defending effects of the diatom, *Amphora coffeaeformis*, as a feed additive versus the deleterious effects (mainly on gonads) caused by microplastics (MPs) in Nile tilapia, *Oreochromis niloticus*. Groups of male tilapia were pre-fed diets with four different supplementation levels of *A. coffeaeformis* (0%, 2.5%, 5%, and 7.5%) for 70 days, then exposed to 10 mg/L MPs for 15 days. Thereafter, samples were taken from the four experimental groups and the control fish group, for evaluating blood picture, erythrocytes alterations, biochemical parameters, catalase (CAT), superoxide dismutase (SDO), and total antioxidant capacity (TAC). In addition, male reproductive performance was assessed by quantifying the follicle-stimulating hormone (FSH), luteinizing hormone (LH), and testosterone (T), also testicular sections and GSI% were also assessed. The results showed that the number of leucocytes were significantly ($p < 0.05$) higher, but the number of red blood cells (RBCs), the level of Hemoglobin (Hb), the percentage of Hematocrit (Ht), the number of platelets, and the eosinophil percentages were significantly decreased ($p < 0.05$) when fish exposed to MPs. Biochemical parameters (ALP, glucose, uric acid, albumin, and A/G ratio) were significantly increased after MP exposure compared with the control group. Furthermore, MPs induced a significant decline in both serum LH and T levels. Testicular, histological, degenerative changes and testis-ova were found in the MP-exposed fish. Thus, *A. coffeaeformis* supplementation displayed ameliorative properties that detoxified the negative effects of MPs. This study provides a better understanding of the reproductive injuries caused by MPs exposure and evidence for the use of *A. coffeaeformis* as a natural remedy in freshwater tilapia.

Keywords Biochemical parameters · Hematological changes · Testis-ova · Oxidative stress · Ameliorative properties

Introduction

In recent years, concerns have been raised about the incorporation of microplastics (MPs) in the aquatic food web. MPs are

synthetic polymers with a diameter of less than 5 mm (Thompson et al. 2009). The increased industrial production of plastic has led to excessive amounts of plastic along coasts and in rivers, seas, and oceans around the world where they accumulate and fragment into smaller particles of microplastics and nanoplastics (Barnes et al. 2009). MPs have a negative effect on aquatic organisms; they can combine with other pollutants, such as plastic additives or organic pollutants, to create more damage (Browne et al. 2013; Hammer et al. 2012). That is why MPs are characterized as the greatest category of pollutants in the world at present (Derraik 2002; Galgani et al. 2013).

Studies have reported that MPs may induce oxidative stress and exert negative effects on the antioxidant defenses of some invertebrates, such as monogonont rotifers (*Brachionus koreanus*), copepods (*Paracyclops nana*), and Chinese mitten crabs (*Eriocheir Sinensis*) (Jeong et al. 2016; Yu et al. 2018). MP-exposed fish could modify antioxidant biomarkers and boost lipid peroxidation in some species, such as

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Highlights

- Pre-feeding with *Amphora coffeaeformis* was investigated against the toxicity of microplastics in *Oreochromis niloticus*.
- This study is the first to find out that microplastics prompt endocrine disruption.
- Testis-ova in male tilapia fish was observed for the first time after MP exposure.

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sheepshead minnows (*Cyprinodon variegatus*) (Choi et al. 2018), red tilapia (*Oreochromis sp.*) (Ding et al. 2018), zebrafish (*Danio rerio*) (Qiao et al. 2019; Wan et al. 2019), and Nile tilapia (*Oreochromis niloticus*) (Hamed et al. 2021; Hamed et al. 2020). In another example, Barboza et al. (2018) found that MPs amplified lipid oxidation in the European sea bass (*Dicentrarchus labrax*) brain, triggering oxidative stress and lipid damage. It was also found that MPs stimulated DNA damage in Nile tilapia (*O. niloticus*) (Hamed et al. 2020). MP exposure has also been found to cause other injurious effects, such as neural dysfunction in gobies (*Pomatoschistus microps*) (Oliveira et al. 2013), hepatic stress in medakas (*Oryzias latipes*) (Rochman et al. 2014), increased mortality in European sea bass (*D. labrax*) (Mazurais et al. 2015), and immune system damage in fathead minnows (*Pimephales promelas*) (Grevén et al. 2016).

Additionally, MPs are a possible cause of fish reproductive stress (Chen et al. 2019), which causes reproductive disruption. However, the stress response in fish gonads caused by MP exposure is still poorly understood. It has been reported that MPs cause endocrine impairment and disrupt the hypothalamus-pituitary-gonadal (HPG) axis and the steroidogenesis pathway (Karami et al. 2016; Rochman et al. 2014; Wan et al. 2019). Furthermore, MPs have been found to delay female gonadal maturation, decrease fecundity (Lönnstedt and Eklöv 2016; Wan et al. 2019), increase the number of reactive oxygen species (ROS) in the gonads of both male and female zebrafish, cause reproductive stress by increasing apoptosis levels and histological alterations in testes (Qiang and Cheng 2021), and decrease the plasma 17 β -estradiol (E2) and testosterone (T) levels in female marine medaka (*Oryzias melastigma*) (Wan et al. 2019).

Recent research has focused on finding effective natural antioxidants to replace synthetic antioxidants currently in use (Glodde et al. 2018; Kumosani et al. 2017; Taghvaei and Jafari 2015). Benthic diatoms are microalgae recognized as natural antioxidants (El-Sayed et al. 2018; Lee et al. 2008). *Amphora coffeaeformis* is a single-celled microalgae present in most aquatic environments that can tolerate various salinities (Sala et al. 1998). Its cell wall is made of opaline silica (Buhmann et al. 2014) and it is an important food source for many invertebrates, fish species, and zooplankton (Kaparapu 2018). Diatoms, such as *A. coffeaeformis*, have a high capability of metal absorption and a high proliferation rate (Anantharaj et al. 2011) that enables them to resist the toxic effects of environmental contaminants (Gaur and Rai 2001). These features support the use of diatoms as antioxidant mediators and detoxifying agents in stress-induced fishes (Sheikhzadeh et al. 2012).

A. coffeaeformis analysis reveals that they are composed of substantial amounts of lipids, proteins, minerals, sugars, carotenoids, chlorophylls, fatty acids, and polyphenols (Chtourou et al. 2015). Moreover, *A. coffeaeformis* has

displayed anti-inflammatory factors (Lauritano et al. 2016), as well as antibacterial (Boukhris et al. 2017), antiviral (Abdel-Wahab 2018), and antioxidant activities that limit the harmful effects of free radicals (El-Sayed et al. 2018; Mekki et al. 2020). *A. coffeaeformis* shows promise as a feed additive, given its bioactive antioxidant constituents that include carotenoids, astaxanthin, canthaxanthin, b-glucans, polyunsaturated fatty acids, sulfated polysaccharides, and vitamins C and E (El-Sayed et al. 2018). As a feed additive, *A. coffeaeformis* in Nile tilapia diets could, therefore, replace antibiotics (Ayoub et al. 2019b) and detoxify and protect against arsenic-induced oxidation as has occurred in the liver tissues of African catfish (*Clarias gariepinus*) (Ayoub et al. 2019a; Moneeb et al. 2020). *A. coffeaeformis* supplementation has been shown to improve fish immunity by reducing pathogenic bacteria and increasing the innate immune system response of fish (Saleh et al. 2020). Additionally, it can enhance male gonadal maturity, increase female fish fecundity (Saleh et al. 2020), and reduce DNA lesions (Karawita et al. 2007).

This study investigated the effect of *A. coffeaeformis* supplementation as pre-feeding on the *Oreochromis niloticus* intoxicated with MPs. This objective was accomplished by the assessment of hematological and biochemical parameters, antioxidant enzymes, hormonal levels, and the reproductive status of male fish. This study will help to elucidate the effects of MP exposure on fish health, with special concerns for gonadal health, and attempt to ameliorate the negative impacts of MPs by supplementing fish diets with *A. coffeaeformis*.

Materials and methods

Experimental design

Nile tilapia fingerlings (360) were obtained from the Baltim experimental farm in the National Institute of Oceanography and Fisheries (NIOF) in Egypt, transferred to the NIOF fish nutrition laboratory, and fed on a commercial diet (25% CP, 4.5 L, Aller Aqua, Egypt) for 1 week to adjust to the experimental conditions. The fish (average body weight, ca. 4 g) were distributed on a triplicate basis into 12 tanks with a 200-L capacity. The water temperature was 27.4 ± 1.2 °C. Oxygen was supplied and kept above 7.0 mg/L. Throughout the experiment period, the natural photoperiod was 11 h of light and 13 h of darkness. The experimental diets contained ascending levels of *A. coffeaeformis* (0%, 2.5%, 5 %, and 7.5%), as described by Saleh et al. (2020) (Table 1). The test diets were given three times a day to satiation. The fish were fed on the experimental diets for 70 days. After this period, ten male fish were selected from each tank, placed in separate aquaria, and kept under MP treatment on a triplicate basis for an extra 15 days. Microplastic particles were added to each

Table 1 Composition and proximate analyses (% DM) of amphora-supplemented diets

Ingredient	Diets (g/kg)				
	A0MP0	A0MP	A1MP	A2MP	A3MP
Fish meal ¹	160	160	160	160	160
Soybean meal ²	350	350	350	350	350
Yellow corn	200	200	200	200	200
Wheat flour	220	220	195	170	145
Sunflower oil	50	50	50	50	50
Vitamins and minerals premix ³	20	20	20	20	20
Amphora meal (g/kg)	0	0	25	50	75
Proximate composition (% DM)					
Crude protein	28.89	28.89	29.50	29.90	29.34
Lipid	7.97	7.97	8.30	8.22	7.99
Ash	7.2	7.2	7.00	6.50	7.10
Fiber	3.66	3.66	3.70	3.90	3.85
NFE ⁴	52.26	52.26	50.50	51.48	51.72
GE (MJ/kg) ⁵	18.34	18.34	19.16	19.20	18.98
Dry matter	90.48	90.48	91.00	89.90	90.89

¹ Lab made (68% protein)² 42% protein³ Vitamins and minerals premix (mg kg⁻¹): p-amino benzoic acid (9.48); D-biotin (0.38); inositol (379.20); niacin (37.92); Ca pantothenate (56.88); pyridoxine HCl (11.38); riboflavin (7.58); thiamine HCl (3.79); L-ascorbyl-2-phosphate Mg (APM) (296.00); folic acid (0.76); cyanocobalamin (0.08); menadione (3.80); vitamin A palmitate (17.85); α-tocopherol (18.96); calciferol (1.14). K₂PO₄ (2.011); Ca₃ (PO₄)₂ (2.736); Mg SO₄·7H₂O (3.058); NaH₂PO₄·2H₂O (0.795)⁴ Nitrogen-free extracts (NFE) = 100 – [% ash + % lipid + % protein + % fiber]⁵ Gross energy was quantified based on 23.6, 39.5, and 17.2 kJ/g for protein, lipid, and carbohydrates, respectively (NRC 2011)

aquarium at a concentration of 10 mg/L, according to Hamed et al. (2019).

The experimental fish groups were including; the control group (A0MP0) where the fish were fed on the control diet and reared in normal conditions, group (A0MP) where the fish were fed on the control diet but exposed to MP-polluted water, and an additional three test groups (A1MP, A2MP, and A3MP) where the fish were reared in MP-polluted water and received increased concentrations of *A. coffeaeformis* (2.5, 5, and 7.5%, respectively) in their diets.

At the end of day 15 of MP exposure, six fish were randomly chosen and sampled from each group. Blood was collected from the caudal vein to measure antioxidant enzymes, hormone levels, and lipid peroxidation. After the fish were anesthetized and weighed separately, they were directly decapitated, and then dissected. The gonads were excised from each fish and weighed. Then, GSI calculations were performed using the following formula: $GSI = \text{gonads weight} /$

gutted weight $\times 100$. Afterwards, the gonads were fixed in neutral buffered formalin (10%).

Fish maintenance and experimental procedures were approved by the Research Committee of the NIOF in Egypt and were in accordance with the Guide for Use and Care of Laboratory Animals (European Communities: Council Directive 2010/63/EU).

Diatom preparation

A. coffeaeformis powder was obtained from the Microalgae Culture Centre at the National Research Centre (NRC) in Egypt. The production procedure was carried out according to El-Sayed et al. (2018). Analysis of the *A. coffeaeformis* powder samples was conducted to determine their biochemical composition (AOAC 1995). They included 26.6% of crude protein, 9.9% of crude lipid, 45.4% of ash, 3.3% of fiber, and 14.8% of total carbohydrate content. The diets were isonitrogenous (ca. 30% CP), isolipidic (ca. 8% L), and isoenergetic (ca. 19 MJ/kg).

Microplastics

MPs consisted of raw powder with irregular-shaped particles. More than 90% of the MPs were less than 100 nm in size. The MP powder was purchased from Toxemerge Pty Ltd. (Melbourne, Australia). A stock solution was prepared from the powder using purified water (Milli-Q) according to the manufacturer's instructions and stored at 4 °C in the dark. The stock solution (2.5 g MP/L) was sonicated before each use. Additional dilutions were prepared immediately from this stock every time the rearing water was changed. The characterization of the MP particles was carried out using light and transmission electron microscopy at the TEMU, Assiut University (JEOL JEM-1200 EX II) (Hamed et al. 2019).

Hematological parameters

Blood samples were collected from the caudal vein of the fish using heparinized syringes and immediately smeared on clean glass slides. Blood smears (six slides from each fish) were dried, fixed in absolute methanol for 1 min, and then stained with hematoxylin and eosin stain (H and E), according to Mekawy et al. (2011). In each slide group, 10,000 cells (minimum of 1,000 cells per slide) were examined under a 40× objective lens with a 10× eyepiece using an OMAX microscope with a 14-MP USB digital camera (CS M837ZFLRC140U) to identify any micronucleated, morphological, or nuclear abnormalities in the red blood cells (RBCs), according to Al-Sabti and Metcalfe (1995). The RBCs and white blood cell (WBCs) counts, hematocrit (HCT), and hemoglobin (HB) were estimated using manual methods. Mean corpuscular volume (MCV), mean corpuscular hemoglobin

(MCH), and mean corpuscular hemoglobin concentration (MCHC) were calculated using the formulae given by Dacie and Lewis (1991).

Biochemical parameters

The remaining blood samples were taken with non-heparinized syringes and then left to clot at 4°C. The coagulated blood samples were centrifuged at 4,000g for 10 min for serum separation. The analyzed biomarkers included aspartate aminotransferase (AST), alanine aminotransferase (ALT), alkaline phosphatase (ALP) activities, glucose (GL), cholesterol, total protein (TP), albumin (AL), globulin (GLO), creatinine (CR), and uric acid (UA). These biomarkers were determined using the test kits from HUMAN GmbH Company, Wiesbaden, Germany.

Antioxidant enzyme activities

Superoxide dismutase (SOD) was evaluated, according to Nishikimi et al. (1972), catalase (CAT) was determined, according to Aebi (1984), and the total antioxidant capacity (TAC) was measured, according to Koracevic et al. (2001).

Hormone measurements

Follicle-stimulating hormone (FSH), luteinizing hormone (LH), and testosterone (T) were measured using ELISA kits (Diagnostics Biochem Canada Inc.; Ontario, Canada). FSH was measured using the test kit (Cat. No. CANFSH-4060) according to Knobil (1980). LH was measured using the test kit (Cat. No. CAN-LH-4040) according to Cumming et al. (1985). T was also measured using the test kit (CAN-TE-250) according to Check et al. (1995). All hormones were measured at 450 nm using an automatic immunodiagnostic analyzer (Sorin Biomedica, Model: 0-2730; S/N = 0654, Chemila SP.A., Italy).

Histological analysis

Fixed testicular samples were dehydrated in ascending grades of ethanol, cleared in methyl benzoate, and embedded in paraffin wax. Five- μ m-thick paraffin sections were cut and stained with H and E stain, and then examined microscopically.

Statistical analysis

Data are expressed as mean $\pm$ standard deviation. Significant differences between the treated and the control groups were tested using the one-way analysis of variance (ANOVA), followed by the Tukey-HSD test for multiple comparisons. The probability of a significant difference was set at $p <$

0.05. The analysis was carried out using the IBM- SPSS package version 23.0.

Results

Hematological parameters

Blood smears from all treated groups had altered erythrocyte shapes, including acanthocytes, crenated cells, teardrop-like cells, and sickle cells. Altered nucleus shapes were also found, including blebbed nuclei, notched nuclei, vacuolated nuclei, and bilobed nuclei (Fig. 1b–f).

The fish group that was exposed to MPs only (A0MP) had a significantly higher number of altered erythrocytes ($p < 0.05$) compared with the other groups (Fig. 2). The group fed on the highest level of *A. coffeaeformis* (A3MP) had a significantly ($p < 0.05$) smaller number of altered erythrocytes compared with the other two *A. coffeaeformis*-fed groups (A1MP and A2MP), but it still displayed a significantly higher ($p < 0.05$) amount than the control group (Fig. 2).

The fish blood analysis results showed a significant decrease ($p < 0.05$) in RBCs, HB, Ht, and platelets in the control group (A0MP0) and the A1MP and A2MP fish groups. In contrast, these blood biomarkers were not significantly different in the A3MP group compared with the control ($p < 0.05$; Table 2). The WBCs were significantly ($p < 0.05$) increased in the A0MP and A1MP groups compared with the A2MP group; however, there were no significant differences in the MCV, MCH, and MCHC percentages among the several treatment groups. WBC differentiation demonstrated a remarkable decrease in the percentage of eosinophils in the A0MP and A1MP groups that coincided with a significant increase in neutrophils ($p < 0.05$). No changes were found in the percentages of monocytes and lymphocytes.

Biochemical parameters

No significant differences ($p < 0.05$) were found in liver enzyme levels (AST and ALT) among the treatment groups (Table 3). ALP levels were the highest in the A0MP fish group, but they remained at control levels in the *A. coffeaeformis*-fed fish group. Serum glucose levels were significantly increased ($p < 0.05$) in the A0MP, A1MP, and A2MP groups compared with the control, but the A3MP group's serum glucose was at control levels. Cholesterol and serum total protein levels were unchanged in the different treatment groups ($p < 0.05$). An obvious elevation in serum uric acid values was recorded in the A0MP, A1MP, and A2MP groups, but the A3MP group's uric acid was at control levels. Uric acid levels displayed a negative correlation with the increasing amounts of *A. coffeaeformis* (Table 3). Serum albumin levels and albumin-to-globulin ratios (A/G) were

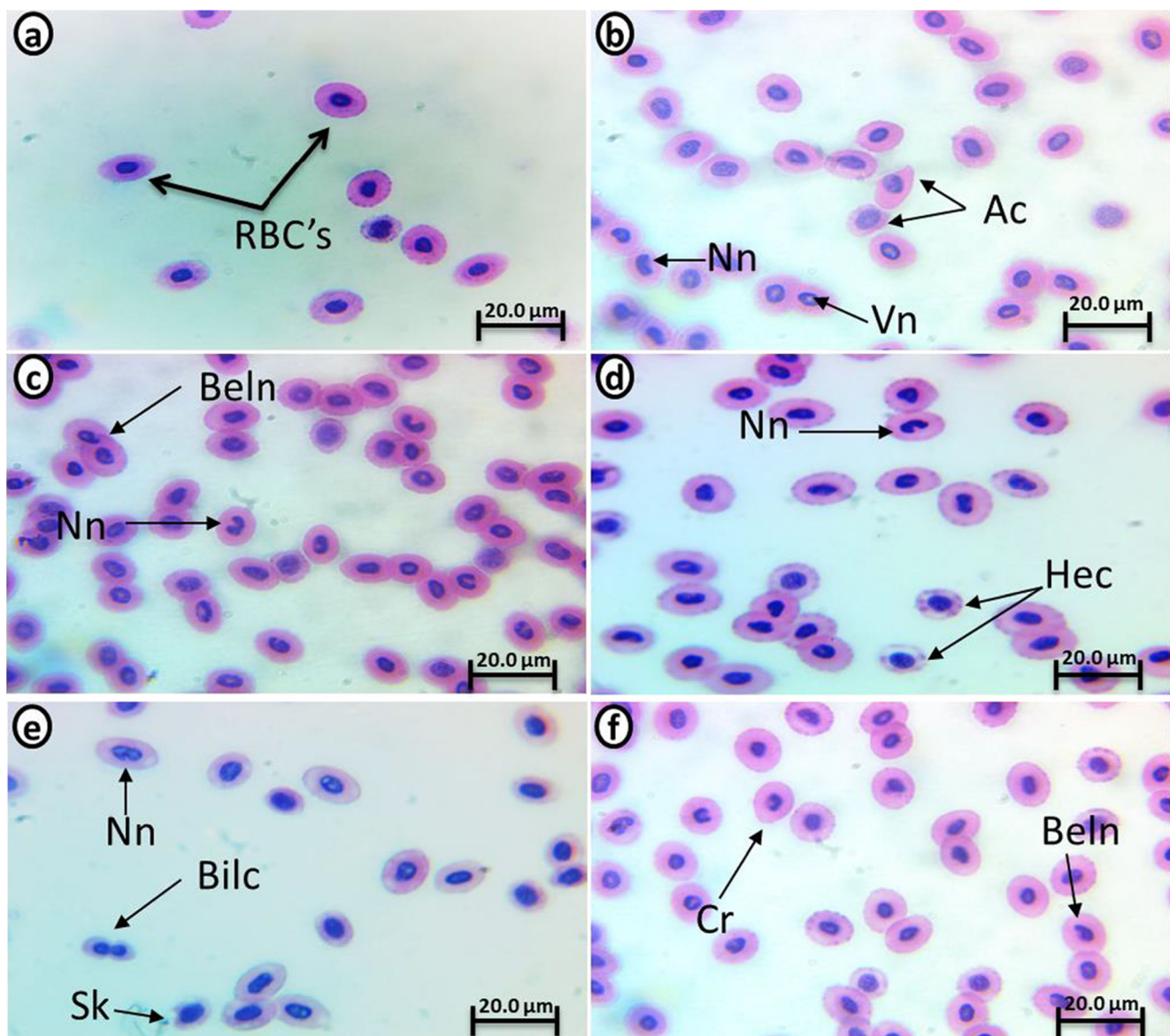


Fig. 1 Representative blood smears from Nile tilapia (*Oreochromis niloticus*). (a) Control (A0MP0); (b, c) (A0MP group) exposed to MPs; (d) A1MP group exposed to MPs + 2.5 % amphora; (e) A2MP group exposed to MPs + 5% amphora; and (f) A3MP exposed to MPs + 7.5%

amphora. RBCs, red blood cells; Ac, acanthocytes; Cr, crenated cells; Beln, blebbed nucleus; Vn, vacuolated nucleus; Hec, hemolyzed cells; Bilc, bilobed cells; Nn, notched nucleus; and Sk, sickle cells. H and E stain was used; scale bar 20 μm .

significantly elevated ($p < 0.05$) in the A0MP and A1MP groups. The A2MP and A3MP groups had the same serum albumin and A/G values as the control.

Antioxidant enzyme activities

TAC levels were unchanged between the A1MP, A2MP, A3MP, and control groups while TAC level was only significantly higher in the A0MP group than the A1MP group (Fig. 3a). A significant elevation ($p < 0.05$) in the serum levels of SOD and CAT enzymes in the A0MP group was observed

compared with the control while the other three fish groups fed on *A. coffeaeformis* had levels comparable with the control group (Fig. 3b, c).

Hormonal activity and GSI

The serum LH and T profiles revealed pronounced suppression in the A0MP group. The LH and T serum levels in the A3MP group were close to those of the control group. The FSH serum levels showed insignificant ($p < 0.05$) changes among the different groups (Fig. 4 a, b, and c). Moreover,

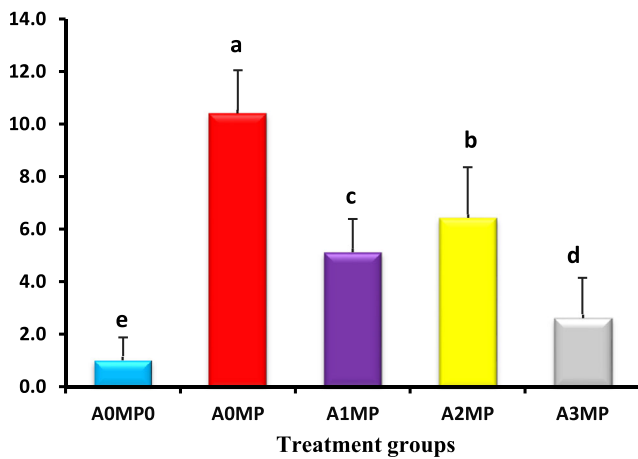


Fig. 2 Erythrocyte alterations (%) of *O. niloticus* fed on different levels of *A. coffeaeformis*-supplemented diets, then exposed to 10 mg/L microplastic. Data are represented as mean $\pm$ SD. Means with different superscript letters (a-e) are significantly different ($p < 0.05$)

GSI displayed significant increases ($p < 0.05$) in the A3MP group (Fig. 4d).

Testicular histology

Light microscopic examination showed that the transverse-sectioned testes from the control group consisted of lobules with different spermatogenic stages separated from each other by interstitial tissue. An accumulation of mature spermatozoa was observed in the lumen of the testicular lobules (Fig. 5a).

The A0MP group had a small number of spermatozoa or no spermatozoa in the lumen of the testicular lobule. Vacuolated seminiferous epithelium and a slight proliferation of the

interstitial connective tissue were also detected in the testes, but the testes still displayed a considerable number of spermatogonia and spermatocytes in the tubule cysts. Other sections of the testes had perinucleolar oocytes inside the testicular lobules (testis-ova) (Fig. 5 b, c, and d).

The A1MP group had proliferated interstitial connective tissue, a marked decrease in the spermatozoa in the lumen of the testicular lobules, and vacuolization of the seminiferous epithelium. The development of a considerable number of early perinucleolar oocytes was found inside the testicular lobules (testis-ova). In addition, degenerative changes in necrotic areas were also observed (Fig. 5e and f).

The A2MP and A3MP groups had no testicular alterations. Sections showed a large number of spermatozoa occupying most of the testicular lobules (Fig. 5g and h).

Discussion

Hematological parameters are the primary indicators of fish health (Thummabancha et al. 2016). In this study, MPs induced erythrocyte alterations in all tested groups compared with the control (A0MP0). Fish groups fed on *A. coffeaeformis*-supplemented diets expressed less significant erythrocyte alterations than the A0MP0 group. A remarkable reduction in the RBCs, Hb, Ht, platelets, and eosinophil percentages after exposure to MPs was detected. Furthermore, the neutrophil percentage was significantly increased after MP exposure. In general, the hematological results indicated that the highest amphora supplementation level (7.5%)

Table 2 Hematological parameters of Nile tilapia (*Oreochromis niloticus*) fed with different *A. coffeaeformis* concentrations, then exposed to 10 mg/L microplastics.

Fish group	A0MP0	A0MP	A1MP	A2MP	A3MP
RBCs (million/mm ³)	2.036 $\pm$ 0.056 ^a	1.813 $\pm$ 0.047 ^b	1.833 $\pm$ 0.047 ^b	1.920 $\pm$ 0.060 ^{ab}	2.023 $\pm$ 0.066 ^a
Hb (g/dL)	8.46 $\pm$ 0.11 ^a	8.00 $\pm$ 0.10 ^b	8.13 $\pm$ 0.25 ^{ab}	7.96 $\pm$ 0.12 ^b	8.33 $\pm$ 0.05 ^{ab}
Ht (PCV) (g/dL)	25.80 $\pm$ 0.34 ^a	23.00 $\pm$ 1.00 ^b	23.33 $\pm$ 0.57 ^b	24.53 $\pm$ 0.23 ^{ab}	25.86 $\pm$ 0.46 ^a
MCV (μ m ³)	126.73 $\pm$ 3.52	126.91 $\pm$ 6.92	127.31 $\pm$ 3.74	133.87 $\pm$ 3.30	127.96 $\pm$ 5.74
MCH (Pg)	41.595 $\pm$ 1.39	44.147 $\pm$ 1.70	44.376 $\pm$ 1.47	43.434 $\pm$ 1.16	41.209 $\pm$ 1.05
MCHC %	32.824 $\pm$ 0.89	34.819 $\pm$ 1.33	34.885 $\pm$ 1.82	32.444 $\pm$ 0.19	32.224 $\pm$ 0.70
Platelets (thousands/mm ³)	315.00 $\pm$ 1.00 ^a	311.33 $\pm$ 0.57 ^b	309.66 $\pm$ 0.57 ^b	310.66 $\pm$ 0.57 ^b	314.00 ^a
WBCs (thousands/mm ³)	850.66 $\pm$ 5.13 ^{ab}	861.66 $\pm$ 5.77 ^a	861.66 $\pm$ 5.13 ^a	846.33 $\pm$ 1.52 ^b	851.33 $\pm$ 6.35 ^{ab}
Lymphocytes %	88.33 $\pm$ 0.57	88.66 $\pm$ 0.57	88.33 $\pm$ 0.57	88.33 $\pm$ 0.57	88.00
Monocytes %	3.66 $\pm$ 0.57	2.66 $\pm$ 0.57	3.00	3.33 $\pm$ 0.57	2.66 $\pm$ 0.57
Neutrophils %	6.00 $\pm$ 0 ^b	7.66 $\pm$ 0.57 ^a	7.66 $\pm$ 0.57 ^a	6.00 $\pm$ 0 ^b	7.33 $\pm$ 0.57 ^a
Eosinophils %	2.00 $\pm$ 0 ^a	1.00 $\pm$ 0 ^b	1.00 $\pm$ 0 ^b	2.33 $\pm$ 0.57 ^a	2.00 $\pm$ 0 ^a

Data are presented as mean $\pm$ SD, n = 6; means with different superscript letters within the same row for each parameter are significantly different ($p < 0.05$)

Table 3 Biochemical parameters (biomarkers) of Nile tilapia (*Oreochromis niloticus*) fed with different *A. coffeaeformis* concentrations, then exposed to 10 mg/L microplastics

Fish group					
Biomarker	A0MP0	A0MP	A1MP	A2MP	A3MP
AST (u/L)	55.00±1.04	53.03±2.02	52.33±1.67	53.26±0.90	54.43±1.00
ALT (u/L)	27.86±0.32	28.03±0.64	27.63±1.59	27.46±0.55	28.36±0.49
ALP (u/L)	24.53±0.05 ^b	27.23±0.28 ^a	26.36±0.55 ^{ab}	25.03±1.33 ^b	25.00±0.52 ^b
Glucose (mg/dL)	97.86±1.27 ^c	111.73±1.15 ^a	114.10±1.53 ^a	105.20±3.34 ^b	98.86±0.05 ^c
Cholesterol (mg/dL)	191.33±5.85	197.33±2.08	191.00±11.26	190.66±3.78	188.33±1.15
Total protein (g/dL)	4.93±0.41	5.26±0.05	5.40±0.20	4.86±0.05	4.86±0.20
uric acid (mg/dL)	12.20±0.17 ^c	13.43±0.05 ^a	13.16±0.25 ^a	12.73±0.05 ^b	12.40±0.17 ^{bc}
Creatinine (mg/dL)	0.53±0.09	0.64±0.02	0.61±0.05	0.55±0.01	0.59±0.07
Albumin (g/dL)	1.03±0.01 ^c	1.27±0.08 ^{ab}	1.41±0.16 ^a	1.09±0.04 ^{bc}	1.12±0.02 ^{bc}
Globulin (g/dL)	2.13±0.01	2.18±0.04	2.17±0.09	2.14±0.05	2.11±0.05
A/G	0.483±0.002 ^c	0.582±0.028 ^{ab}	0.647±0.071 ^a	0.511±0.009 ^{bc}	0.532±0.008 ^{bc}

Data are presented as mean ± SD, n = 6, means with different superscript letters within the same row for each parameter are significantly different ($p < 0.05$). Abbreviations: A/G, albumin-globulin ratio; ALT, alanine amino-transferase; AST, aspartate aminotransferase

ameliorates the effects of MPs since they maintained the hematological parameters at control levels. These results concur with those of Hamed et al. (2019) who found that MP exposure induced hematological deterioration in juvenile Nile tilapia. Moreover, the immune system efficiency is affected by MP uptake that disturbs the organism's defense and health (Espinosa et al. 2017).

Serum biochemical parameters are good indicators of the animals' general health status and can provide early signs of serious changes in stressed organisms (Folmar 1993; Garima and Himanshu 2015; Jacobson-Kram and Keller 2001). It has been reported that MP exposure can induce changes in the biochemical parameters of various fish species, including common goby (*Pomatoschistus microps*) (Oliveira et al. 2013), common carp (*Cyprinus carpio*) (Haghi and Banaee 2017), and Nile tilapia (Hamed et al. 2019). In general, the different serum biomarkers act as indicators of organ dysfunction and the impairment of biological processes. In this study, some serum biomarkers (ALP, glucose, uric acid, albumin, and A/G ratio) were significantly increased due to MP exposure. These biomarkers were not significantly changed and their levels were kept the same as control levels when *A. coffeaeformis* supplementation was used with 5% and 7.5% diets.

The results of this study are in agreement with those of Mekki et al. (2020) who recorded that *A. coffeaeformis* dietary supplementation improved the arsenic-induced hematological parameters in African catfish (*Clarias gariepinus*). Additionally, they are consistent with those of El-Sayed et al. (2018) who proved that the dietary supplementation of *A. coffeaeformis* removed the elements that caused liver tissue damage and was induced by paracetamol in rats. The ameliorative effect of *A. coffeaeformis* may be attributed to its

bioactive components that have many health benefits (Mekki et al. 2020; Saleh et al. 2020) that protect organisms against stressors. Furthermore, Ayoub et al. (2019a) found that *A. coffeaeformis* supplementation in fish diets could increase disease resistance and improve the hematological and biochemical parameters in Nile tilapia.

Antioxidant biomarkers are used to identify the effects of multiple environmental stressors on aquatic organisms (Hook et al. 2014). In this study, SOD and CAT levels were significantly elevated attributable to MP exposure. However, TAC levels were close to control levels in all tested groups, but they were significantly different between the A2MP and A0MP groups. The disruption of antioxidant enzyme activities triggered by MP exposure has been recorded in some studied aquatic organisms (Prinz and Korez 2020) and fish, including common goby (Luis et al. 2015), zebrafish (Lu et al. 2016), sheepshead minnows (Choi et al. 2018), and red tilapia (Ding et al. 2018). The present results are inconsistent with Karami et al. (2016) who found that MPs had no effects on the antioxidant enzyme activities in zebrafish. Changes in CAT and SOD enzymes were limited by *A. coffeaeformis* supplementation. These results agree with El-Sayed et al. (2018) and Moneeb et al. (2020) who suggested that *A. coffeaeformis* extracts could restore antioxidant enzymes to control levels. *A. coffeaeformis* has antioxidative properties and is considered a natural antioxidant (Lee et al. 2008; Lee et al. 2009) which is able to stimulate a specific set of biochemical and physiological actions to repel the toxic effects of environmental pollutants (Gaur and Rai 2001).

Fish reproductive success has a long-term effect on the population constancy and the integrity of the entire aquatic ecosystem. The current results demonstrated that exposure to MP pollution could negatively impact the fertility of male

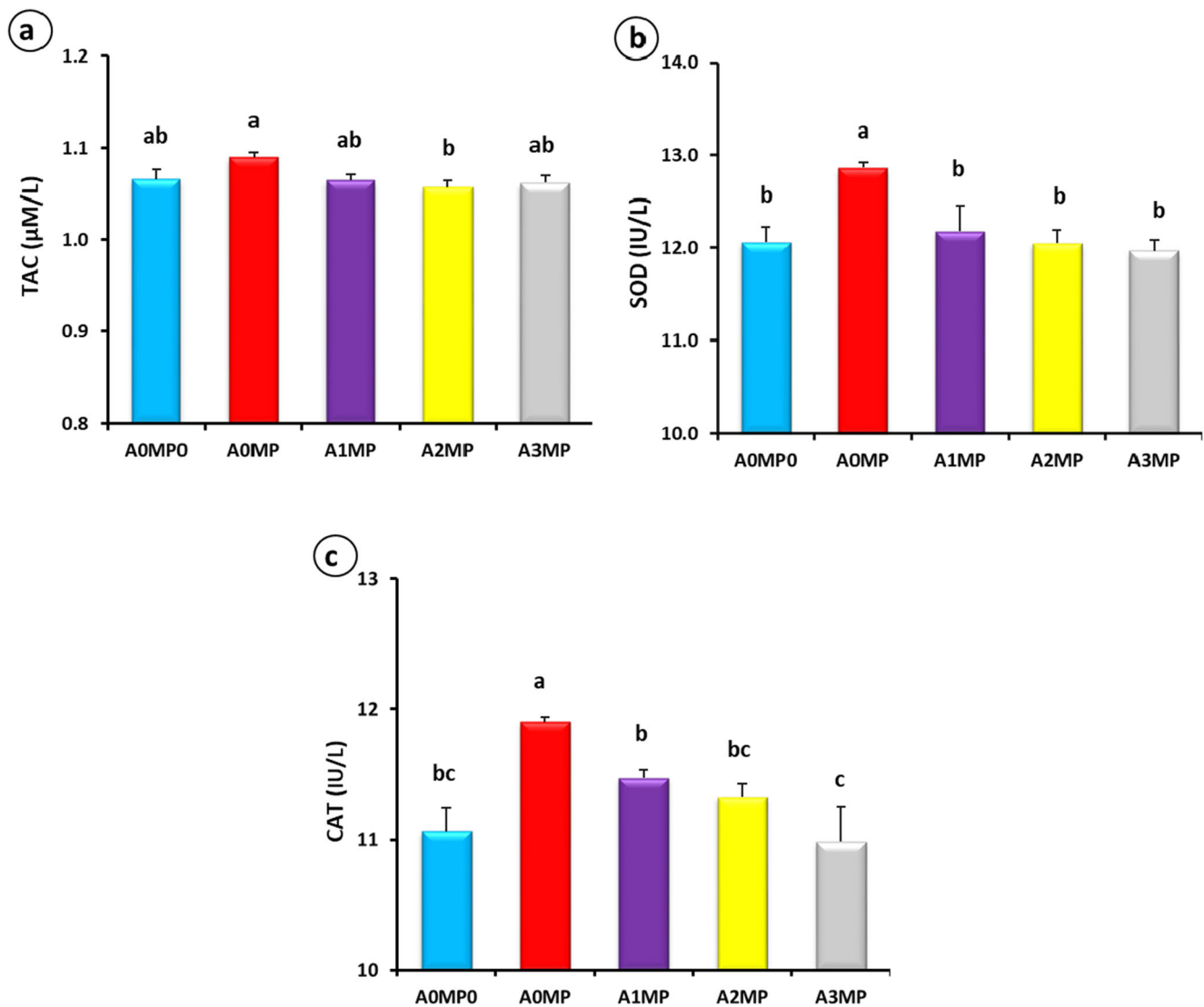


Fig. 3 Antioxidant enzyme activity (TAC, SOD, and CAT) in Nile tilapia (*Oreochromis niloticus*) fed with different *A. coffeaeformis* concentrations, then exposed to 10 mg/L microplastics. Data are

presented as mean $\pm$ SD. N = 6. Means with different superscript letters (a-c) are significantly different ($p < 0.05$)

tilapia and induce oxidative stress. The potential negative impacts of MPs were reduced in fish fed on higher *A. coffeaeformis* levels.

It is suggested that MPs act indirectly to the hypothalamus-pituitary axis to alter the synthesis and secretion of gonadotropin that leads to the interruption of sex steroid production and testicular degeneration. The alterations seen in this study have also been recorded in male marine medaka (*Oryzias melastigma*) and may be explained by the up-regulation of gene expression in genes involved in the HPG axis and the steroidogenesis pathway (Wan et al. 2019). MPs also down-regulated the transcription of gonadotropin-releasing hormone (GnRH) in the hypothalamus of African catfish (*Clarias gariepinus*) (Karami et al. 2016) and the vitellogenin and

choriogenin in the liver of male Japanese medaka (*Oryzias latipes*) (Rochman et al. 2014). In accordance with the histological alterations described in this study, Rochman et al. (2014) defined the abnormal growth of male germ cells and suggested that this abnormal growth may be due to small clusters of oogonia that undergo atrophy when male Japanese medaka were exposed to environmentally relevant MP concentrations. Wang et al. (2019) explained different histological alterations in testicular tissues after MP exposure, including an increase in the interstitial tissue, a disorder of the arrangement of seminiferous lobules, the dissolution of the basal membrane, and a loose arrangement of spermatocytes. Moreover, a reduction in the testis basement membrane thickness was also recorded (Qiang and Cheng 2021). In contrast to

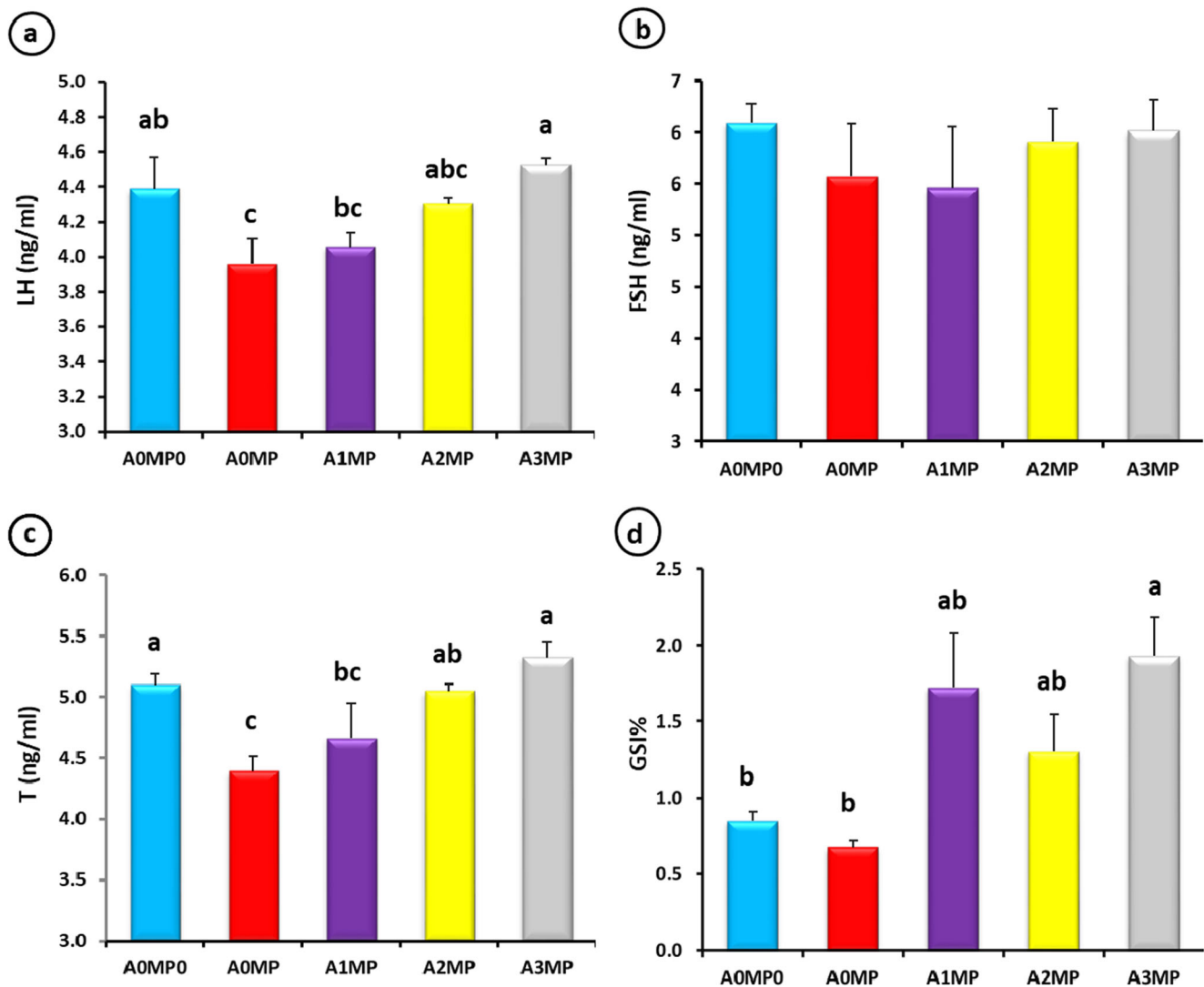


Fig. 4 Hormonal activity (LH, FSH, and T) and GSI% value in Nile tilapia (*Oreochromis niloticus*) fed with different *A. coffeaeformis* concentrations, and then exposed to 10 mg/L microplastics. Data are

presented as mean ± SD. n = 6. Means with different superscript letters (a-c) are significantly different ($p < 0.05$)

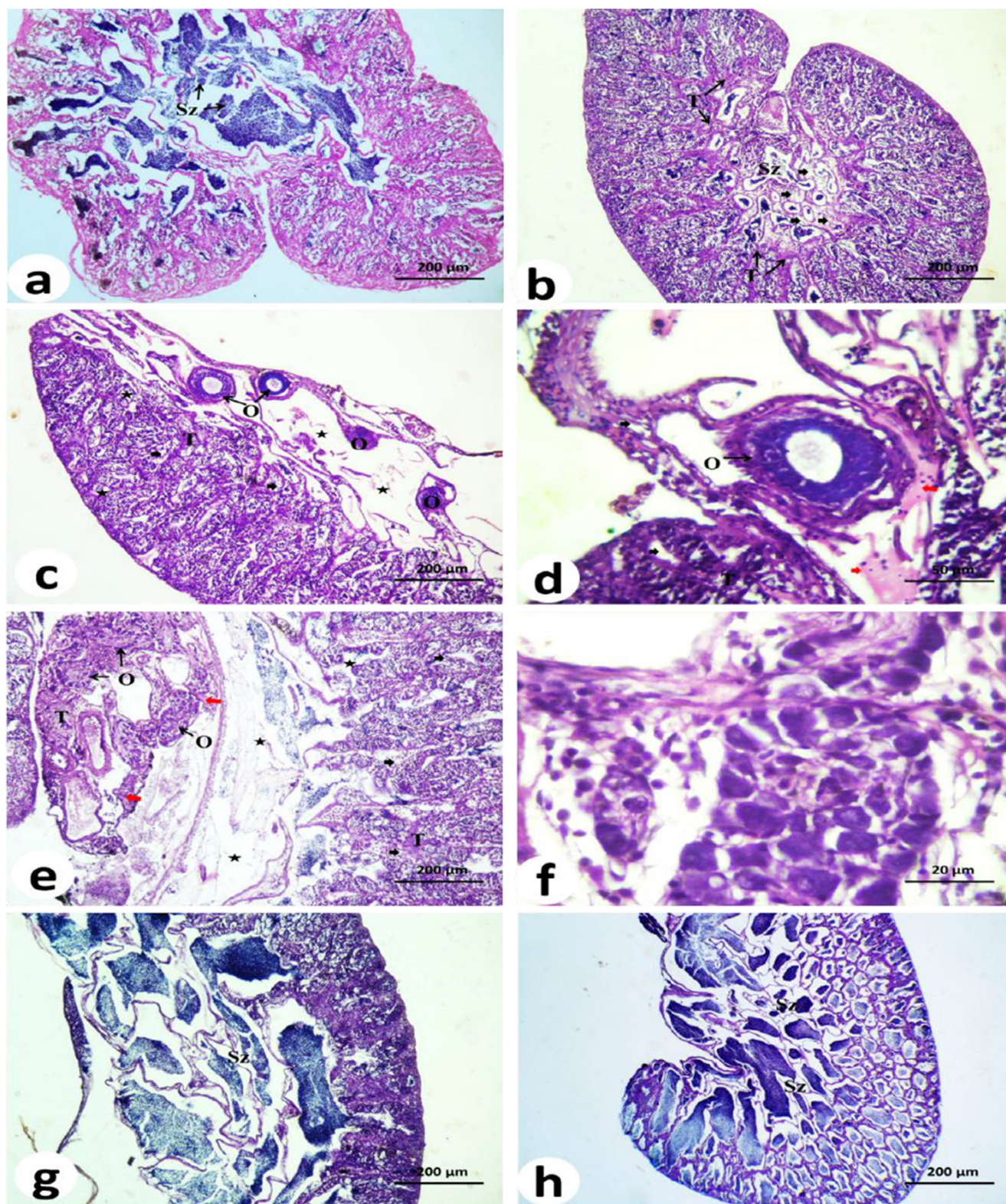
the present results, MP exposure significantly increased the levels of T in male marine medaka (Wan et al. 2019).

This study indicated that the testis' histological alterations found in fish exposed to MPs were the most improved. This improvement was due to the two higher doses of *A. coffeaeformis* supplementation in the A2MP group (5%) and the A3MP group (7.5%) that kept the LH and T levels the same as in control fish and improved the GSI values.

Furthermore, the current results suggest that MP exposure indirectly reduces the reproductive performance in Nile tilapia through oxidative stress and this comes in agreement with Wang et al. (2019). In addition, Qiang and Cheng (2021) proposed that the overall stress

of detoxification and the induction of antioxidant enzyme activities are contributing factors to reproductive impairments. Gonads are most predisposed to oxidative stress because gonads display a higher frequency of cell division, mitochondrial oxygen intake, and unsaturated fatty acids compared with other tissues (Asadi et al. 2017; Ciani et al. 2015). It has been previously shown that stressors, such as chemical exposure, could induce testicular oxidative stress, cause apoptosis in germ cells, and negatively affect spermatogenesis (Asadi et al. 2017).

In conclusion, fish exposed to 10 mg/L of MPs for 15 days induced hematological and biochemical changes in



O. niloticus. Furthermore, MPs prompted oxidative stress, hormonal disruption, and testicular damage and induced

testis-ova in male tilapia. The highest tested supplementation level of *A. coffeaeformis* (A3MP; 7.5%) prevented oxidative

Fig. 5 Microscopic pictures of *O. niloticus* testes after 15 days of microplastic exposure: (a) shows the cystic arrangement in control testis with different spermatogenic cells and spermatozoa (SZ) in the lumen of the cysts. (b–d) show sections of testis from AOMP with different alterations, a proliferation of interstitial tissue (T), vacuolization of seminiferous epithelium (arrowheads), and a small number of spermatozoa (SZ) or a lack of spermatozoa in the lumen of the testicular lobules (asterisks). Different oocyte developmental stages were seen inside the testicular lobule (O) and necrosis (red arrow) was also present. (e) Testis sections from A1MP with a proliferation of the interstitial tissue (T), vacuolization of the seminiferous epithelium (arrowheads), a lack of spermatozoa from the lumen of the testicular lobules (asterisks), and necrosis (red arrow). A nest of early perinucleus oocytes (O) was also present to form the testes-ova. (f) Magnification of the early perinucleus oocytes. (g) Section from A2MP showed normal cystic arrangement with a fair number of mature spermatozoa (SZ). (h) Section from A3MP with cysts occupied by mature spermatozoa (SZ). H and E staining was used.

stress, testicular damage, and negative hematological alterations. The present results suggest that *Amphora* feeding could be used as a supporting agent against plastic pollution in the aquaculture sector and aquaponics.

Availability of data and materials All data generated or analyzed during this study are included in the research article.

Author contribution Experimental design: RFI, NES, AHS. Experiment and analysis: RFI, NES, AHS. Data interpretation: RFI, NES, AHS. Writing and revision: RFI, NES, AHS. All authors read and approved the final manuscript.

Declarations

Ethical approval It is not applicable.

Competing interests The authors declare no competing interests.

References

- Abdel-Wahab AM (2018) In-vitro studies on antiviral effects of *Galaxaura elongata* marine algae on white spot syndrome virus. *Benha Vet Med J* 34:162–171. <https://doi.org/10.21608/bvmj.2018.54231>
- Aebi H (1984) Catalase in vitro. *Methods Enzymol* 105:121–126
- Al-Sabti K, Metcalfe CD (1995) Fish micronuclei for assessing genotoxicity in water. *Mutat Res* 343:121–135
- Anantharaj K, Govindasamy C, Natanamurugaraj G, Jeyachandran S (2011) Effect of Heavy metals on marine diatom *amphora coffeaeformis* (Agardh. Kutz). *Glob J Environ Res* 5:112–117
- AOAC (1995) Official methods of analysis of AOAC International. 6th edn. Association of Official Analytical Chemists, Arlington, VA, USA pp 1141
- Asadi N, Bahmani M, Kheradmand A, Rafieian-Kopaei M (2017) The impact of oxidative stress on testicular function and the role of antioxidants in improving it: a review. *J Clin Diagn Res: JCDR* 11:1e01–1e05
- Ayoub HF, Abdelghany MF, El-Sayed AB (2019a) Effects of diatoms *Amphora coffeaeformis* on growth parameters, nonspecific immunity and protection of the Nile tilapia (*Oreochromis niloticus*) to *Aeromonas hydrophila* infection. *Egyptian J Aquat Biol Fish* 23:413–426
- Ayoub HF, Abdelghany MF, El-Sayed B, Abo El-Khair A (2019b) Effects of diatoms *Amphora coffeaeformis* on growth parameters, non specific immunity and protection of the Nile tilapia (*Oreochromis niloticus*) to *Aeromonas hydrophila* infection. *Egypt J Aquat Biol Fish* 23:413–426
- Barboza LGA, Vieira LR, Guilhermino L (2018) Single and combined effects of microplastics and mercury on juveniles of the European seabass (*Dicentrarchus labrax*): changes in behavioural responses and reduction of swimming velocity and resistance time. *Environ Pollut (Barking, Essex : 1987)* 236:1014–1019
- Barnes DK, Galgani F, Thompson RC, Barlaz M (2009) Accumulation and fragmentation of plastic debris in global environments. *Phil Trans R Soc B* 364:1985–1998
- Boukhris S, Athmouni K, Hamza-Mnif I, Siala-Elleuch R, Ayadi H, Nasri M, Sellami-Kamoun A (2017) The potential of a brown microalga cultivated in high salt medium for the production of high-value compounds. *Biomed Res Int* 2017:4018562
- Browne MA, Niven SJ, Galloway TS, Rowland SJ, Thompson RC (2013) Microplastic moves pollutants and additives to worms, reducing functions linked to health and biodiversity. *Curr Biol* 23:2388–2392
- Buhmann MT, Poulsen N, Klemm J, Kennedy MR, Sherrill CD, Kröger N (2014) A tyrosine-rich cell surface protein in the diatom *Amphora coffeaeformis* identified through transcriptome analysis and genetic transformation. *PLoS One* 9:e110369
- Check JH, Ubelacker L, Lauer CC (1995) Falsely elevated steroidal assay levels related to heterophile antibodies against various animal species. *Gynecol Obstet Investig* 40:139–140
- Chen Q, Allgeier A, Yin D, Hollert H (2019) Leaching of endocrine disrupting chemicals from marine microplastics and mesoplastics under common life stress conditions. *Environ Int* 130:104938
- Choi JS, Jung YJ, Hong NH, Hong SH, Park JW (2018) Toxicological effects of irregularly shaped and spherical microplastics in a marine teleost, the sheepshead minnow (*Cyprinodon variegatus*). *Mar Pollut Bull* 129:231–240
- Chtourou H, Dahmen I, Jebali A, Karray F, Hassairi I, Abdelkafi S, Ayadi H, Sayadi S, Dhoubi A (2015) Characterization of *Amphora* sp., a newly isolated diatom wild strain, potentially usable for biodiesel production. *Bioprocess Biosyst Eng* 38:1381–1392
- Ciani F, Cocchia N, d'Angelo D, Simona T (2015) Influence of ROS on ovarian functions
- Cumming DC, Vickovic MM, Wall SR, Fluker MR (1985) Defects in pulsatile LH release in normally menstruating runners. *J Clin Endocrinol Metab* 60:810–812
- Dacie JV, Lewis SM (1991) Practical haematology. Churchill Livingstone, Edinburgh; New York
- Derraik JGB (2002) The pollution of the marine environment by plastic debris: a review. *Mar Pollut Bull* 44:842–852
- Ding J, Zhang S, Razanajatovo RM, Zou H, Zhu W (2018) Accumulation, tissue distribution, and biochemical effects of polystyrene microplastics in the freshwater fish red tilapia (*Oreochromis niloticus*). *Environ Pollut (Barking, Essex : 1987)* 238:1–9
- El-Sayed AEB, Aboulthana WM, El-Feky AM, Ibrahim NE, Seif MM (2018) Bio and phyto-chemical effect of *Amphora coffeaeformis* extract against hepatic injury induced by paracetamol in rats. *Mol Biol Rep* 45:2007–2023
- Espinosa C, Cuesta A, Esteban MÁ (2017) Effects of dietary polyvinylchloride microparticles on general health, immune status and expression of several genes related to stress in gilthead seabream (*Sparus aurata* L.). *Fish Shellfish Immunol* 68:251–259. <https://doi.org/10.1016/j.fsi.2017.07.006>

- Folmar LC (1993) Effects of chemical contaminants on blood chemistry of teleost fish: a bibliography and synopsis of selected effects. *Environ Toxicol Chem* 12:337–375
- Galgani F, Hanke G, Werner S, De Vrees L (2013) Marine litter within the European Marine Strategy Framework Directive. *ICES J Mar Sci* 70:1055–1064
- Garima P, Himanshu P (2015) Impact of heavy metal, arsenic trioxide on biochemical profile of teleost, *Clarias batrachus* (Linn.). *Int J of Life Sciences* 3:141–146
- Gaur JP, Rai LC (2001) “Heavy metal tolerance in algae,” in *Algal Adaptation to Environmental Stresses*. In Rai Gaur JP (ed) Physiological, Biochemical and Molecular. Springer-Verlag, Berlin, pp 363–388. https://doi.org/10.1007/978-3-642-59491-5_12
- Glodde F, Günel M, Kinsel ME, AbuGhazaleh A (2018) Effects of natural antioxidants on the stability of omega-3 fatty acids in dog food. *J Vet Res* 62:103–108. <https://doi.org/10.1515/jvetres-2018-0014>
- Greven A-C, Merk T, Karagöz F, Mohr K, Klapper M, Jovanovi B, Palj D (2016) Polycarbonate and polystyrene nanoplastic particles act as stressors to the innate immune system of fathead minnow (*Pimephales promelas*). *Environ Toxicol Chem* 35(12):3093–3100
- Haghi B, Banaee M (2017) Effects of micro-plastic particles on paraquat toxicity to common carp (*Cyprinus carpio*): biochemical changes. *Int J Environ Sci Technol* 14:521–530. <https://doi.org/10.1007/s13762-016-1171-4>
- Hamed M, Soliman HAM, Osman A, Sayed A (2019) Assessment the effect of exposure to microplastics in Nile Tilapia (*Oreochromis niloticus*) early juvenile: I. blood biomarkers. *Chemosphere* 228: 345–350
- Hamed M, Soliman HAM, Osman A, Sayed AE-DH (2020) Antioxidants and molecular damage in Nile Tilapia (*Oreochromis niloticus*) after exposure to microplastics. *Environ Sci Pollut Res Int* 27:14581–14588
- Hamed M, Soliman HAM, Badrey AEA, Osman AGM (2021) Microplastics induced histopathological lesions in some tissues of tilapia (*Oreochromis niloticus*) early juveniles. *Tissue Cell* 71: 101512
- Hammer J, Kraak MH, Parsons JR (2012) Plastics in the marine environment: the dark side of a modern gift. *Rev Environ Contam Toxicol* 220:1–44
- Hook S, Gallagher E, Batley G (2014) The role of biomarkers in the assessment of aquatic ecosystem health. *Integr Environ Assess Manag* 10(3):327–341
- Keller KA (2006) Developmental and reproductive toxicology. In: Jacobson-Kram D, Keller KA, editors. *Toxicology Testing Handbook: Principles, Applications and Data Interpretation*. Marcel Dekker, New York, pp. 195–252
- Jeong C-B, Won E, Kang H-m, Lee M, Hwang D-S, Hwang U-K, Zhou B, Souissi S, Lee S, Lee J-S (2016) Microplastic size-dependent toxicity, oxidative stress induction, and p-JNK and p-p38 activation in the monogonont rotifer (*Brachionus koreanus*). *Environ Sci Technol* 50(16):8849–8857
- Kaprapu J (2018) Application of Microalgae in Aquaculture. *Phykos* 48(1):21–26
- Karami A, Romano N, Galloway T, Hamzah H (2016) Virgin microplastics cause toxicity and modulate the impacts of phenanthrene on biomarker responses in African catfish (*Clarias gariepinus*). *Environ Res* 151:58–70
- Karawita R, Senevirathne M, Athukorala Y, Affan A, Lee Y-j, Kim S, Lee J-B, Jeon Y (2007) Protective effect of enzymatic extracts from microalgae against DNA damage induced by H₂O₂. *Mar Biotechnol* 9:479–490
- Knobil E (1980) The neuroendocrine control of the menstrual cycle. *Recent Prog Horm Res* 36:53–88
- Koracevic D, Koracevic G, Djordjević V, Andrejevic S, Cosic V (2001) Method for the measurement of antioxidant activity in human. *J Clin Pathol* 54:356–361
- Kumosani T, Balamash K, Ghashlan H, Ahmed Y, Baothman O, Zeyadi M et al (2017) Potential antioxidant and anti-proliferative activities of biologically active marine algae extracts. *J Pharm Res Int* 19:1–7. <https://doi.org/10.9734/jpri/2017/38089>
- Lauritano C, Andersen JH, Hansen E, Albrigtsen M, Escalera L, Esposito F et al (2016) Bioactivity screening of microalgae for antioxidant, antiinflammatory, anticancer, anti-diabetes, and antibacterial activities. *Front Mar Sci* 3:68. <https://doi.org/10.3389/fmars.2016.00068>
- Lee S-H, Karawita R, Affan A, Lee J-B, Lee B-j, Jeon Y (2008) Potential antioxidant activities of enzymatic digests from benthic diatoms *Achnanthes longipes*, *Amphora coffeaeformis*, and *Navicula* sp. (Bacillariophyceae). *J Food Sc Nutr* 13:166–175
- Lee S-H, Karawita R, Affan A, Lee J-B, Lee K-w, Lee B-j, D-w K, Jeon Y (2009) Potential of benthic diatoms *Achnanthes longipes*, *Amphora coffeaeformis* and *Navicula* sp. (Bacillariophyceae) as antioxidant sources. *Algae* 24:47–55
- Lönnstedt OM, Eklöv P (2016) Environmentally relevant concentrations of microplastic particles influence larval fish ecology. *Science* 352: 1213–1216
- Lu Y, Zhang Y, Deng Y, Jiang W, Zhao Y, Geng J-j, Ding L, Ren H-q (2016) Uptake and accumulation of polystyrene microplastics in zebrafish (*Danio rerio*) and toxic effects in liver. *Environ Sci Technol* 50(7):4054–4060
- Luis LG, Ferreira P, Fonte E, Oliveira M, Guilhermino L (2015) Does the presence of microplastics influence the acute toxicity of chromium(VI) to early juveniles of the common goby (*Pomatoschistus microps*)? A study with juveniles from two wild estuarine populations. *Aquat Toxicol* 164:163–174
- Mazurais D, Ernande B, Quazuguel P, Severe A, Huelvan C, Madec L, Mouchel O, Soudant P, Robbens J, Huvet A, Zambonino-Infante J (2015) Evaluation of the impact of polyethylene microbeads ingestion in European sea bass (*Dicentrarchus labrax*) larvae. *Mar Environ Res* 112:78–85
- Mekkawy IAA, Mahmoud U, Sayed A (2011) Effects of 4-nonylphenol on blood cells of the African catfish *Clarias gariepinus* (Burchell, 1822). *Tissue Cell* 43(4):223–229
- Mekkawy IAA, Mahmoud U, Moneeb RH, Sayed AE-DH (2020) Significance assessment of *Amphora coffeaeformis* in arsenic-induced hemato-biochemical alterations of African catfish (*Clarias gariepinus*). *Frontiers in Marine Science*
- Moneeb RH, Mekkawy IA, Mahmoud UM, Sayed AE-DH (2020) Histopathological and ultrastructure studies on hepatotoxicity of arsenic in *Clarias gariepinus* (Burchell, 1822): hepatoprotective effect of *Amphora coffeaeformis*. *Scientific African* 8:e00448
- Nishikimi M, Appaji N, Yagi K (1972) The occurrence of superoxide anion in the reaction of reduced phenazine methosulfate and molecular oxygen. *Biochem Biophys Res Commun* 46(2):849–854
- NRC (2011) Nutrient requirements of fish and shrimp. The National Academies Press, Washington, DC, 392 pp
- Oliveira M, Ribeiro A, Hylland K, Guilhermino L (2013) Single and combined effects of microplastics and pyrene on juveniles (0+ group) of the common goby *Pomatoschistus microps* (Teleostei, Gobiidae). *Ecol Indic* 34:641–647
- Prinz N, Korez p (2020) Understanding how microplastics affect marine biota on the cellular level is important for assessing ecosystem function: a review
- Qiang L, Cheng J (2021) Exposure to polystyrene microplastics impairs gonads of zebrafish (*Danio rerio*). *Chemosphere* 263:128161
- Qiao R, Sheng C, Lu Y, Zhang Y, Ren H, Lemos B (2019) Microplastics induce intestinal inflammation, oxidative stress, and disorders of metabolome and microbiome in zebrafish. *Sci Total Environ* 662: 246–253
- Rochman C, Kurobe T, Flores I, Teh SJ (2014) Early warning signs of endocrine disruption in adult fish from the ingestion of polyethylene with and without sorbed chemical pollutants from the marine environment. *Sci Total Environ* 493:656–661

- Sala S, Sar E, Ferrario M (1998) Review of materials reported as containing *Amphora coffeaeformis* (Agardh) Kützinger in Argentina. *Diatom Research* 13:323–336
- Saleh NE, Ismail RF, Sayed AH, Zaghloul EH, Saleh H (2020) Comprehensive assessment of benthic diatom (*Amphora coffeaeformis*) as a feed additive in Nile tilapia (*Oreochromis niloticus*) diet. *Aquac Res* 51:3506–3519
- Sheikhzadeh N, Tayefi-Nasrabadi H, Khani Oushani A, Najafi Enferadi MH (2012) Effects of *Haematococcus pluvialis* supplementation on antioxidant system and metabolism in rainbow trout (*Oncorhynchus mykiss*). *Fish Physiol Biochem* 38:413–419. <https://doi.org/10.1007/s10695-011-9519-7>
- Taghvaei M, Jafari SM (2015) Application and stability of natural antioxidants in edible oils in order to substitute synthetic additives. *J Food Sci Technol* 52:1272–1282. <https://doi.org/10.1007/s13197-013-1080-1>
- Thompson R, Moore CJ, Saal FS, Swan S (2009) Plastics, the environment and human health: current consensus and future trends. *Phil Trans R Soc B* 364:153–2166. <https://doi.org/10.1098/rstb.2009.0053>
- Thummabancha K, Onpam N, Srisapoom P (2016) Analysis of hematologic alterations, immune responses and metallothionein gene expression in Nile tilapia (*Oreochromis niloticus*) exposed to silver nanoparticles. *J Immunotoxicol* 13:909–917
- Wan Z, Wang C, Zhou J, Shen M, Wang X, Fu Z, Jin Y (2019) Effects of polystyrene microplastics on the composition of the microbiome and metabolism in larval zebrafish. *Chemosphere* 217:646–658
- Wang J, Li Y, Lu L, Zheng M, Zhang X, Tian H, Wang W, Ru S (2019) Polystyrene microplastics cause tissue damages, sexspecific reproductive disruption and transgenerational effects in marine medaka (*Oryzias melastigma*). *Environ Pollut* 254:113024. <https://doi.org/10.1016/j.envpol.2019.113024>
- Yu P, Liu Z, Wu D, Chen M, Lv W, Zhao Y (2018) Accumulation of polystyrene microplastics in juvenile *Eriocheir sinensis* and oxidative stress effects in the liver. *Aquat Toxicol* (Amsterdam, Netherlands) 200:28–36

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